Source Systems, Data Ingestion, and Pipelines

DeepLearning.AI

Week 3

DeepLearning.AI

DataOps

Week 3 Overview

DataOps

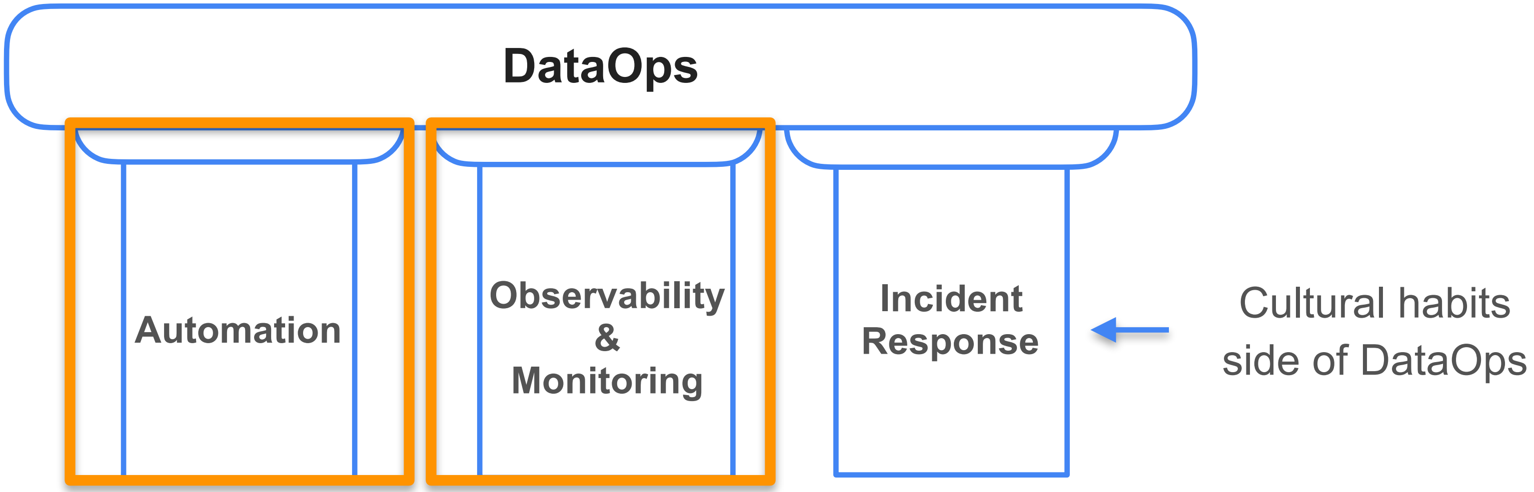
DataOps

Set of practices and cultural habits centered around building robust data systems and delivering high quality data products.

DevOps

Set of practices and cultural habits that allow software engineers to efficiently deliver and maintain high quality software products.

The Three Pillars of DataOps



Automation

Continuous Integration and Continuous Delivery (CI/CD)

Build → Test → Integrate → Deploy

Infrastructure as code

Code that acts to deploy the resources required to run your data pipeline

AWS CloudFormation

HashiCorp Terraform

Week 3 Labs

HashiCorp Terraform

Hands-on experience creating resources using Terraform

Monitoring Data Quality & other Observability Metrics

Interviews with Industry Experts

Chris Bergh

Barr Moses

Abe Gong

Chad Sanderson

DataKitchen

MC MONTE CARLO

great expectations

gable

DataOps

Chris Bergh
CEO at DataKitchen

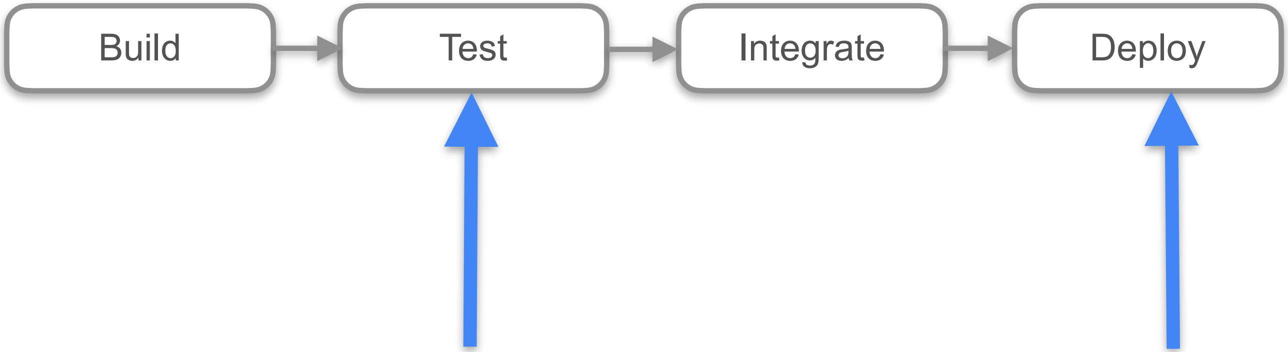
DeepLearning.AI

DataOps - Automation

DataOps Automation

DevOps Automation

Continuous Integration and Continuous Delivery (CI/CD)

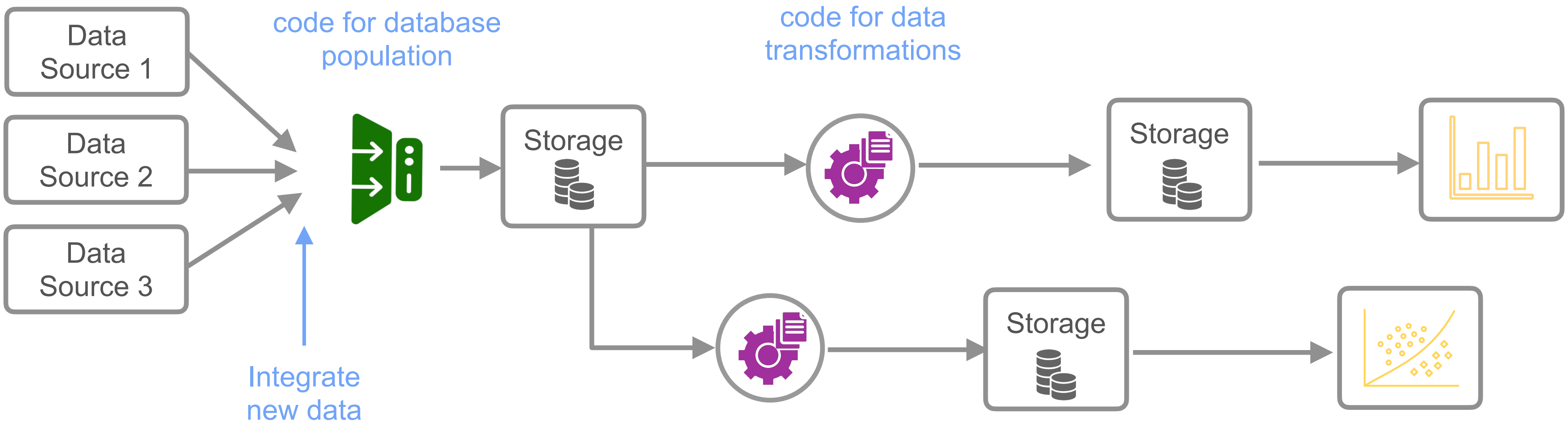


Automatic review and testing of new code

Automatic delivery or deployment into production

DataOps Automation

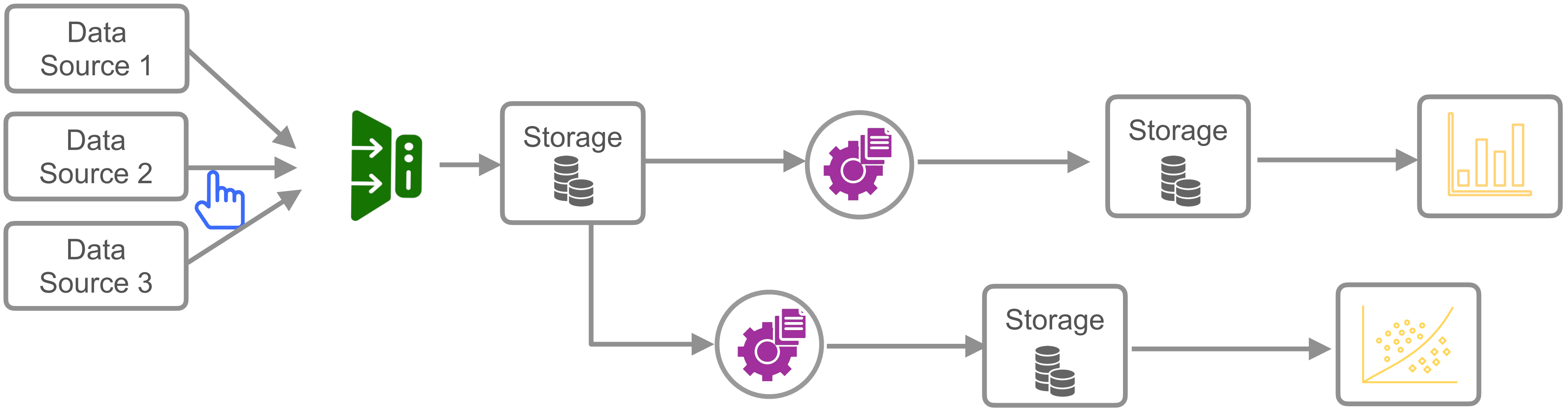
CI/CD can be applied directly to code and data within your data pipeline



DataOps Automation

Automation of data pipeline running

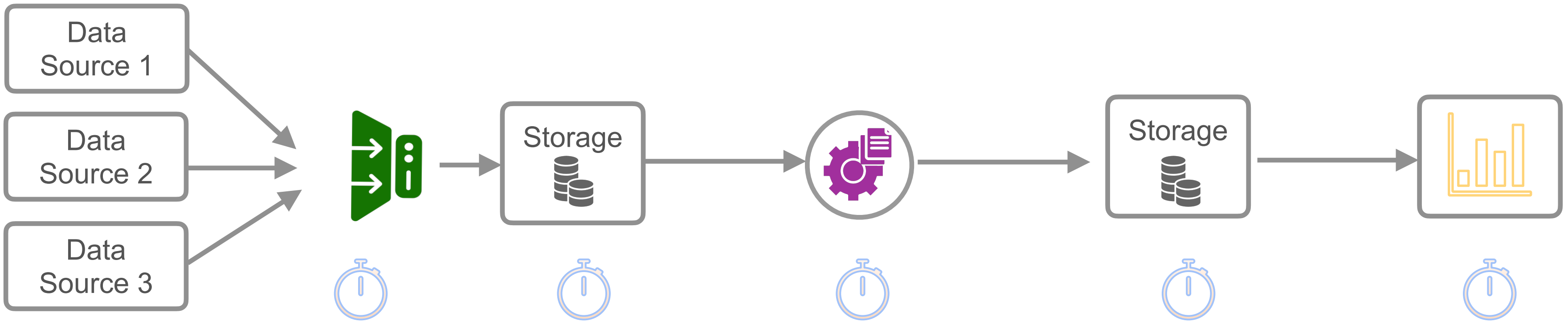
No automation: run all processes manually



DataOps Automation

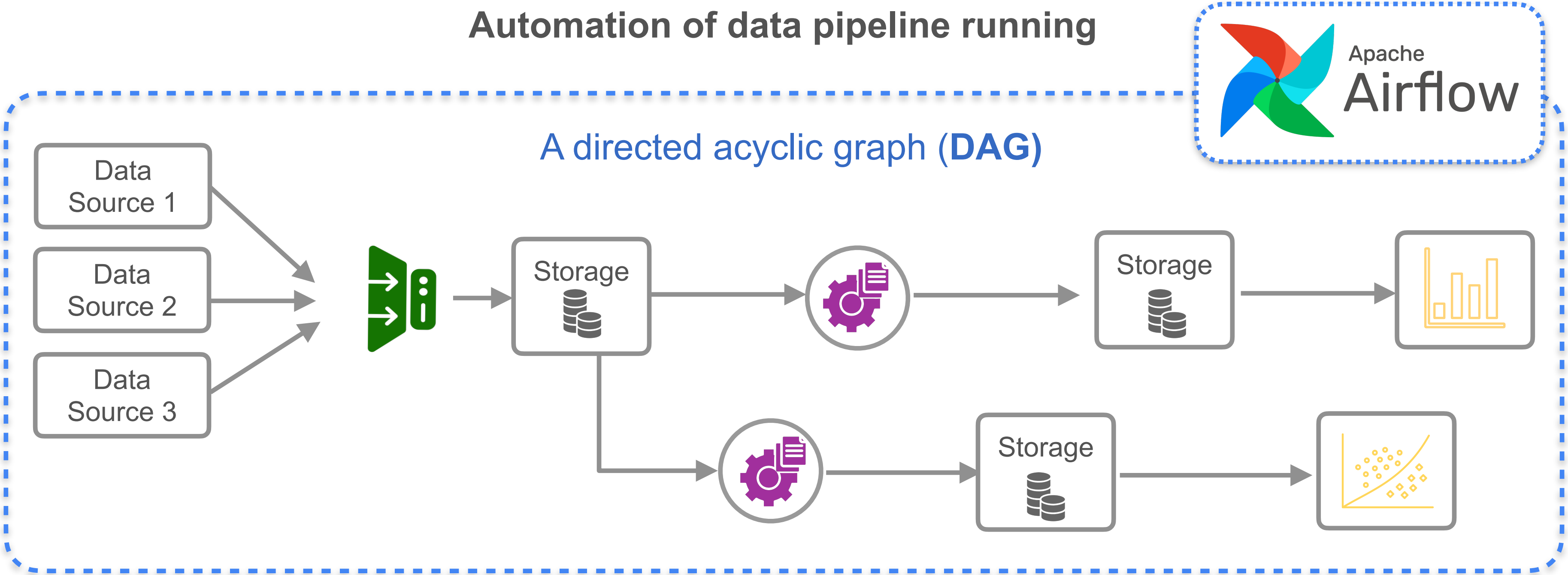
Automation of data pipeline running

Pure scheduling: run stages of your pipeline according to a schedule



DataOps Automation

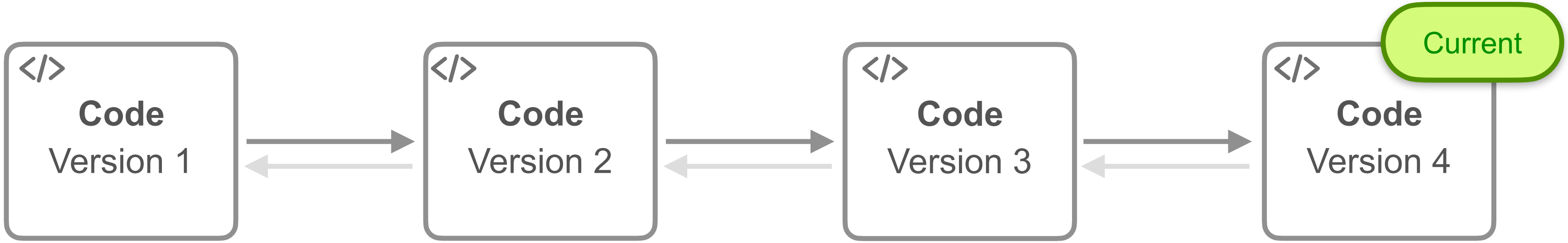
Automation of data pipeline running



DataOps Automation

Key Underpinning CI/CD : Version control

Track changes in the **code**



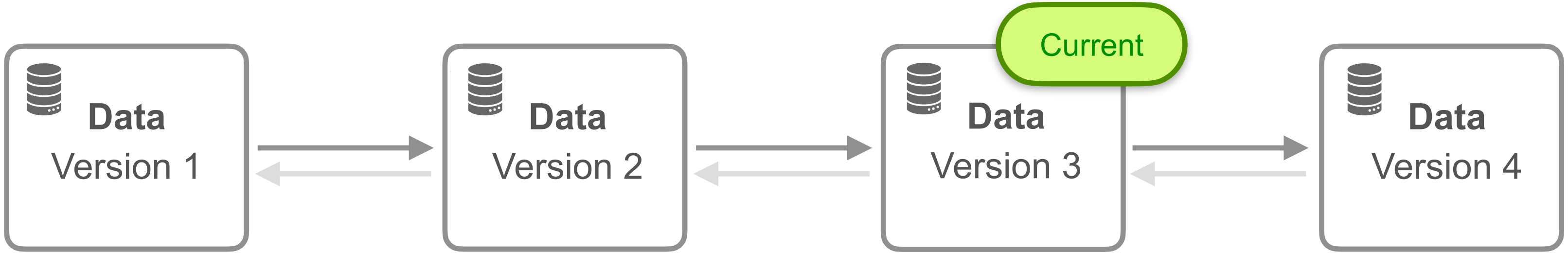
GitHub

Revert to a previous version

DataOps Automation

Key Underpinning CI/CD : Version control

Track changes in the **code** and the **data**

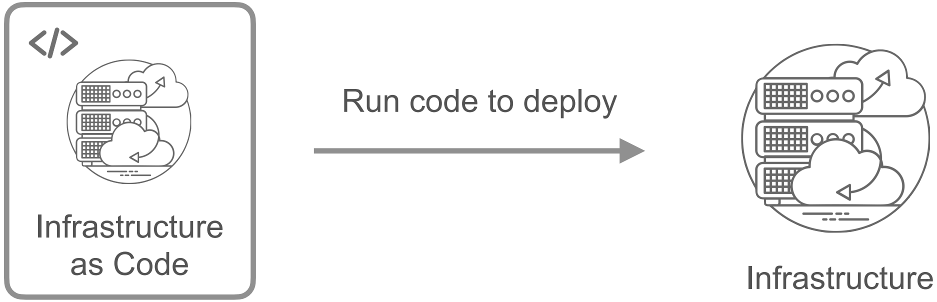


Revert to previous version

DataOps Automation

Key Automation Aspect: Infrastructure as Code

Maintain the design of your infrastructure as a codebase

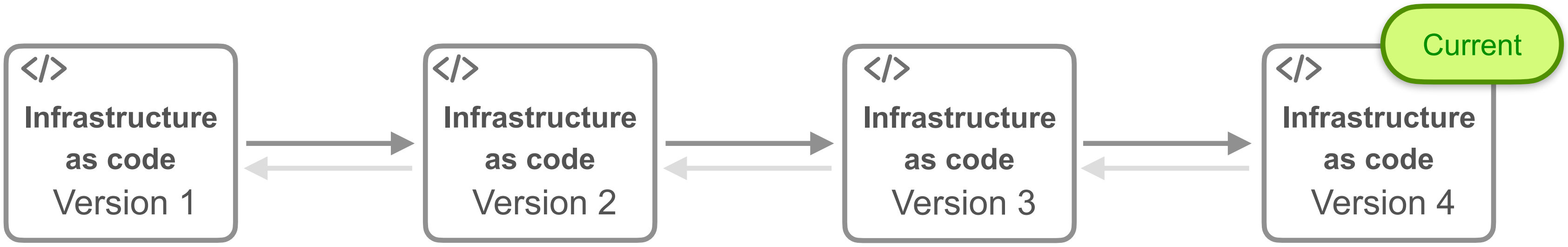


Modify this code to redefine your infrastructure

DataOps Automation

Version control over your entire infrastructure

Track changes in the **code** defining your infrastructure



Revert to a previous version

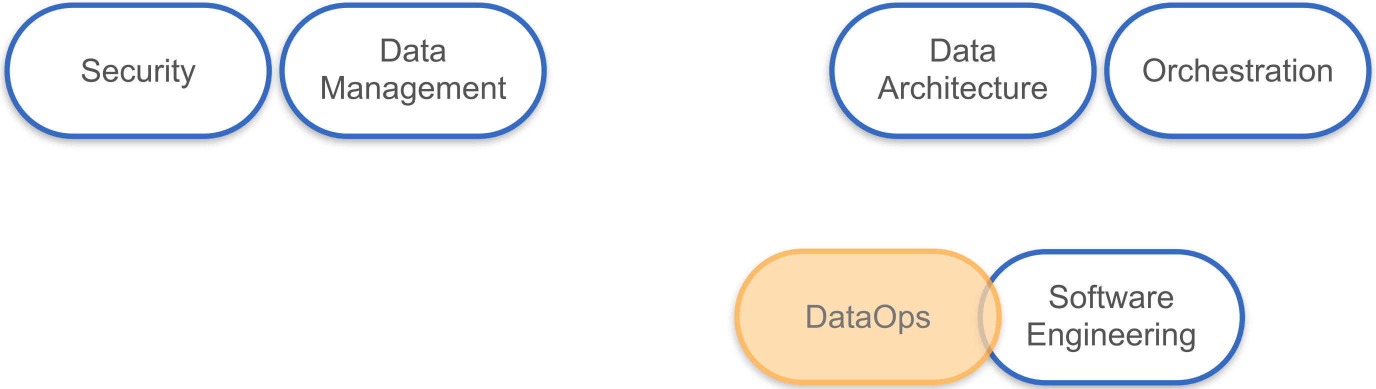
DataOps

The Undercurrents

Security
Data Management
DataOps
Data Architecture
Orchestration
Software Engineering

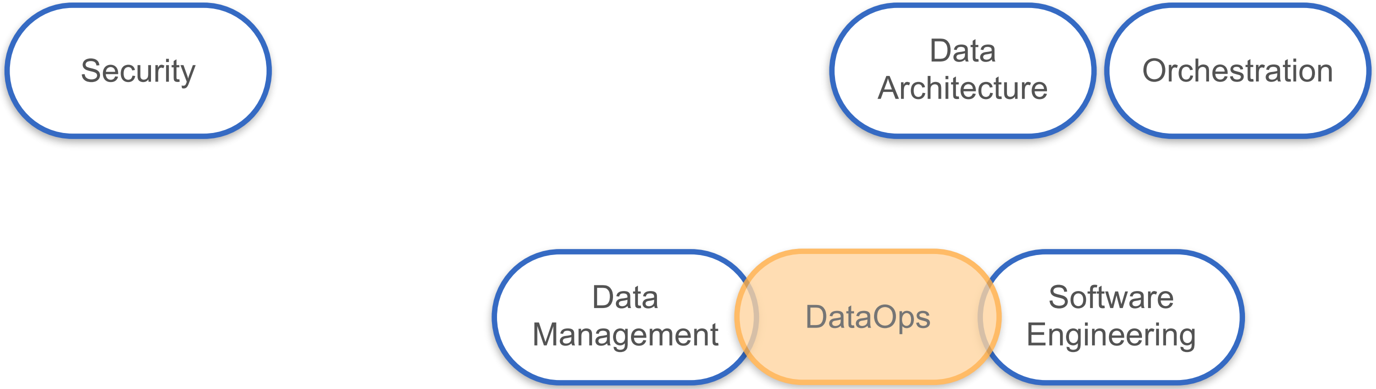
DataOps

The Undercurrents



DataOps

The Undercurrents



DeepLearning.AI

DataOps - Automation

Infrastructure as Code

Infrastructure as Code

1970s

Configuration Management

Challenge : configuration of a series of physical machines

BASH
THE BOURNE-AGAIN SHELL

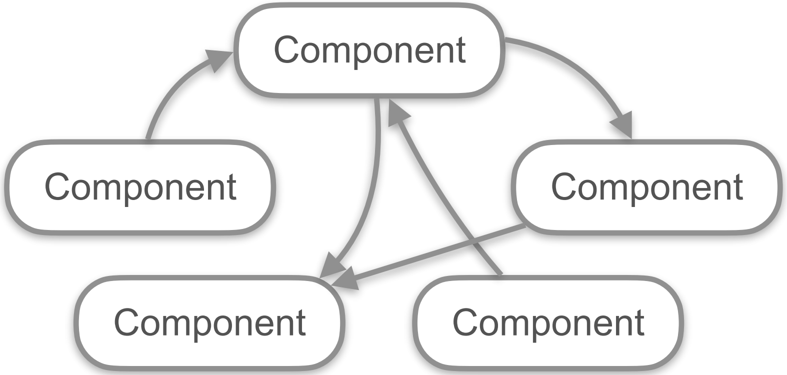
To automate some configuration tasks

Infrastructure as Code

1970s — Configuration Management

2006

Amazon Elastic Compute Cloud (Amazon EC2)



Infrastructure as Code

1970s Configuration Management

2006 Amazon Elastic Compute Cloud (Amazon EC2)

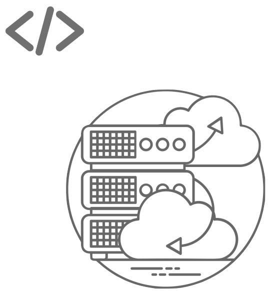
2010s **Development of Infrastructure as Code Tools**

HashiCorp Terraform

AWS CloudFormation

ANSIBLE

Infrastructure as Code



Infrastructure as Code

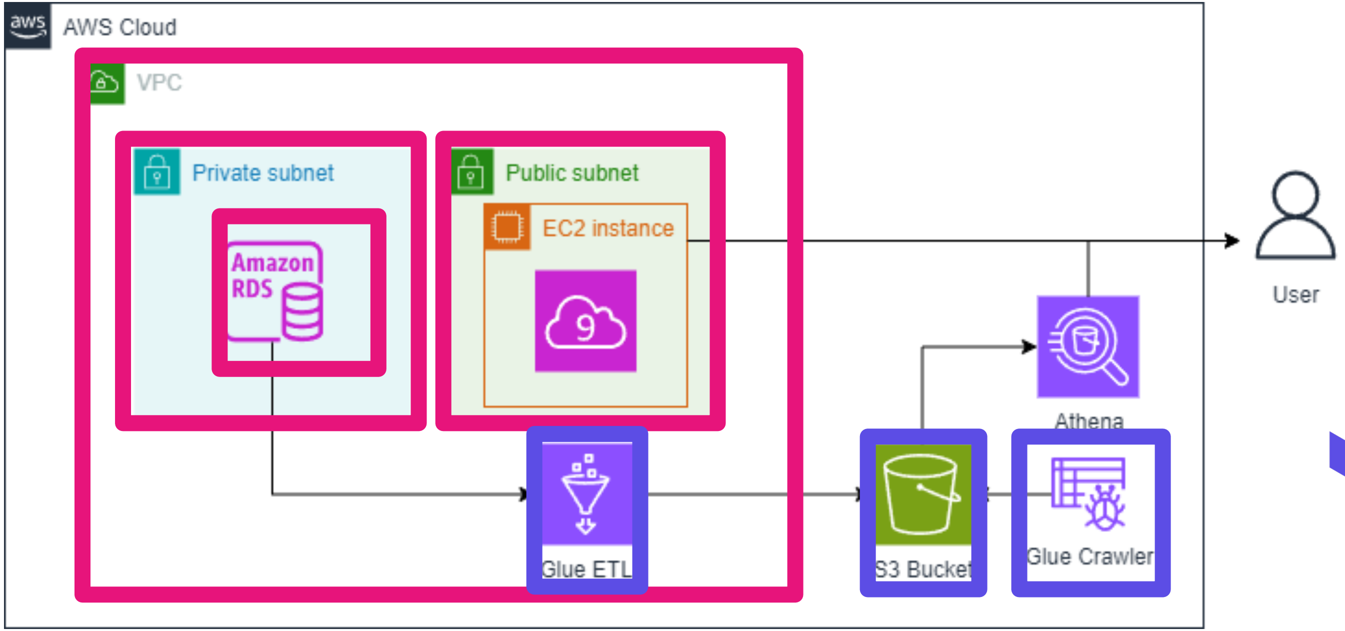
AWS CloudFormation

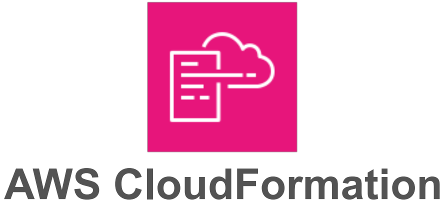
HashiCorp Terraform

ANSIBLE

- Manage infrastructure resources on the cloud using code
- No manual clicking or writing bash scripts

Course 1 Week 2 Lab





Infrastructure as Code Tools

HashiCorp Terraform

- Used in the lab
- Well supported with great documentation

AWS CloudFormation

- Native to AWS
- Well supported with great documentation

Terraform Configuration Files (C1W2 Lab)

```
resource "aws_s3_bucket" "data_lake" {
  bucket_prefix = "${var.project}-datalake-${data.aws_caller_identity.current.account_id}-"
}

resource "aws_s3_bucket_public_access_block" "data_lake" {
  bucket = aws_s3_bucket.data_lake.id

  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls      = true
  restrict_public_buckets = true
}

resource "aws_s3_bucket" "scripts" {
  bucket_prefix = "${var.project}-scripts-${data.aws_caller_identity.current.account_id}-"
}

resource "aws_s3_bucket_public_access_block" "scripts" {
  bucket = aws_s3_bucket.scripts.id

  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls      = true
  restrict_public_buckets = true
}

resource "aws_s3_object" "glue_job_script" {
  bucket = aws_s3_bucket.scripts.id
```

```
resource "aws_glue_catalog_database" "analytics_database" {
  name        = "${var.project}-analytics-db"
  description = "Database for performing analytics on OLTP data"
}

resource "aws_glue_connection" "rds_connection" {
  name = "${var.project}-rds-connection"

  connection_properties = {
    JDBC_CONNECTION_URL = "jdbc:mysql://${var.host}:${var.port}/${var.database}"
    USERNAME            = var.username
    PASSWORD            = var.password
  }

  physical_connection_requirements {
    availability_zone      = data.aws_subnet.private_a.availability_zone
    security_group_id_list = [data.aws_security_group.db_sg.id]
    subnet_id              = data.aws_subnet.private_a.id
  }
}

resource "aws_glue_crawler" "s3_crawler" {
  name          = "${var.project}-analytics-db-crawler"
  database_name = aws_glue_catalog_database.analytics_database.name
  role          = aws_iam_role.glue_role.arn

  s3_target {
    path = "s3://${aws_s3_bucket.data_lake.bucket}/gold"
  }
```

Terraform Configuration File (s3.tf)

Keyword

Resource Type

Name of the resource

```
resource "aws_s3_bucket" "data_lake" {
  bucket_prefix = "${var.project}-datalake-${data.aws_caller_identity.current.account_id}-"
}

resource "aws_s3_bucket_public_access_block" "data_lake" {
  bucket = aws_s3_bucket.data_lake.id

  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls      = true
  restrict_public_buckets = true
}
```

Key-value pairs

1. Set up the S3 bucket
2. Configure the bucket and provide public access to it

Terraform Language

```
resource "aws_s3_bucket" "data_lake" {
  bucket_prefix = "${var.project}-datalake-${data.aws_caller_identity.current.account_id}-"
}

resource "aws_s3_bucket_public_access_block" "data_lake" {
  bucket = aws_s3_bucket.data_lake.id

  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls      = true
  restrict_public_buckets = true
}

resource "aws_s3_bucket" "scripts" {
  bucket_prefix = "${var.project}-scripts-${data.aws_caller_identity.current.account_id}-"
}

resource "aws_s3_bucket_public_access_block" "scripts" {
  bucket = aws_s3_bucket.scripts.id

  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls      = true
  restrict_public_buckets = true
}

resource "aws_s3_object" "glue_job_script" {
  bucket = aws_s3_bucket.scripts.id
```

Domain-Specific Language

HCL (HashiCorp Configuration Language)

The HCL Syntax

```
resource "aws_vpc" "main" {
  cidr_block       = "10.0.0.0/16"
  instance_tenancy = "default"

  tags = {
    Name = "main"
  }
}
```

```
resource "aws_instance" "web" {
  ami           = data.aws_ami.ubuntu.id
  instance_type = "t3.micro"

  tags = {
    Name = "HelloWorld"
  }
}
```

Create or update a VPC

Virtual private cloud (VPC)

Create and Update an EC2 instance

Instance

The HCL Syntax

```
resource "google_compute_instance" "default" {
  name         = "my-instance"
  machine_type = "n2-standard-2"
  zone         = "us-central1-a"

  tags = ["foo", "bar"]

  boot_disk {
    initialize_params {
      image = "debian-cloud/debian-11"
      labels = {
        my_label = "value"
      }
    }
  }
```

Provision a GCP (Google Cloud Platform) compute instance

Terraform Language

HCL is a declarative language

You just have to declare what you want the infrastructure to look like

```
resource "aws_s3_bucket" "data_lake" {
  bucket_prefix = "${var.project}-datalake-${data.aws_caller_identity.current.account_id}-"
}

resource "aws_s3_bucket_public_access_block" "data_lake" {
  bucket = aws_s3_bucket.data_lake.id

  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls      = true
  restrict_public_buckets = true
}
```

S3 Bucket

s3.tf

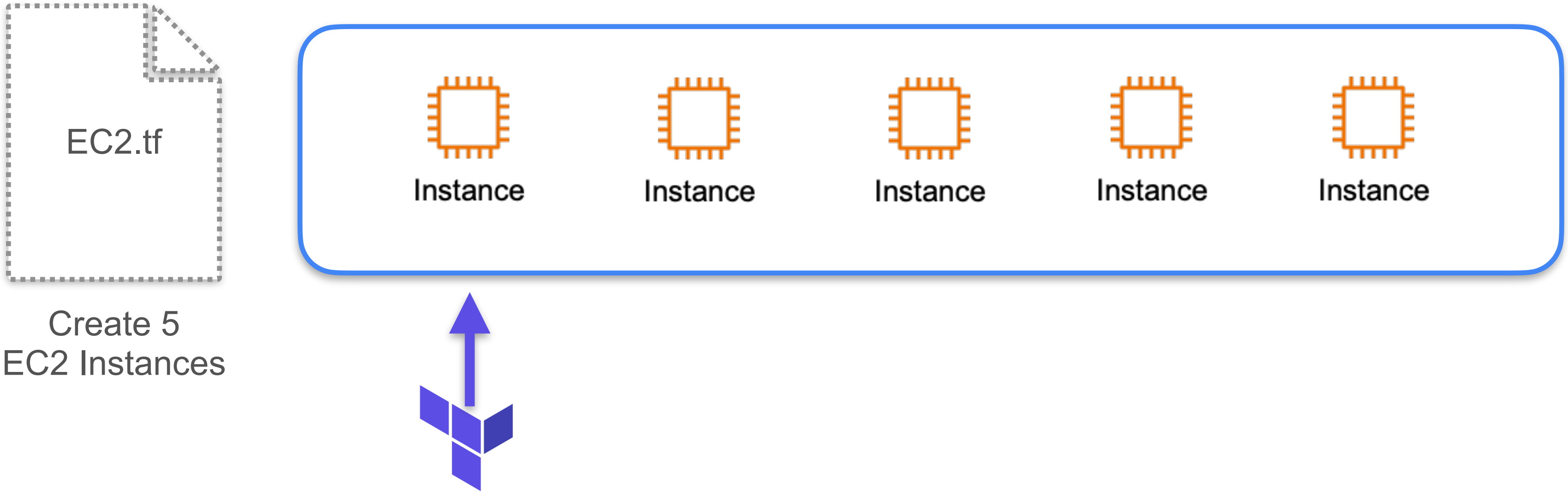
The desired end-state of the infrastructure:

resources and their configurations

Terraform is Highly idempotent!

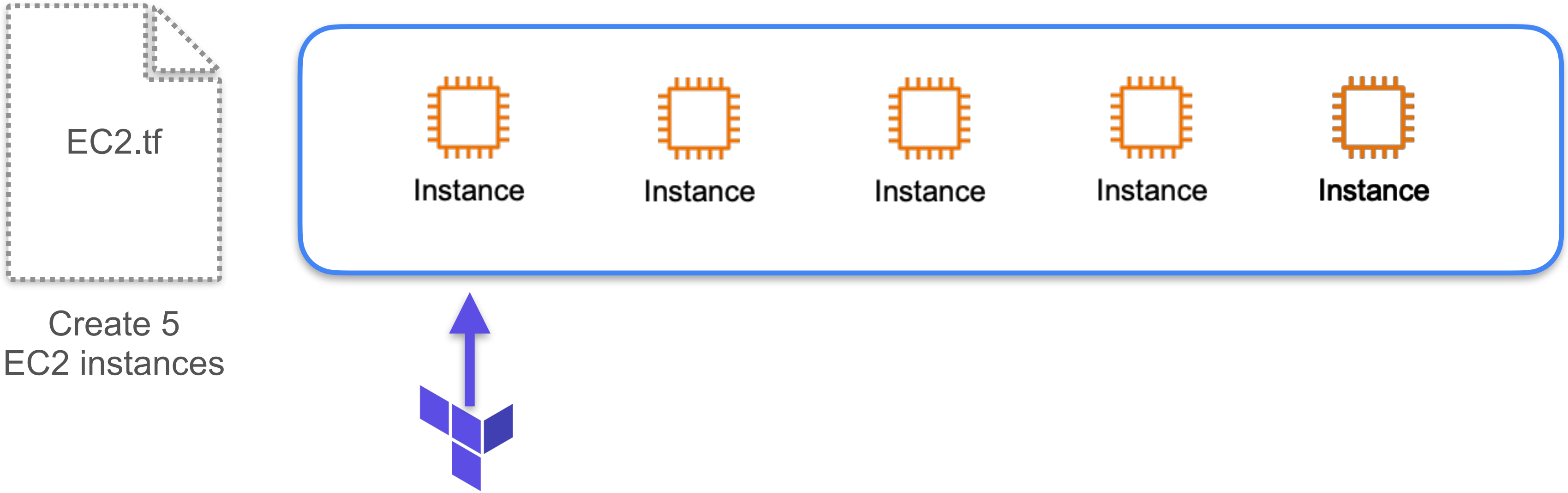
Terraform Language

HCL is a declarative language



Terraform Language

HCL is a declarative language



Create 5
EC2 instances

Imperative/Procedural Language

BASH
THE BOURNE-AGAIN SHELL

Imperative: specify the exact sequence of configuration tasks.

DeepLearning.AI

DataOps - Automation

Terraform - Creating an EC2 instance

Terraform

You write the configuration files to define your resources

Terraform prepares your workspace

1. Installs necessary files
2. Creates an execution plan

Terraform

You write the configuration files to define your resources

↓

Terraform prepares your workspace

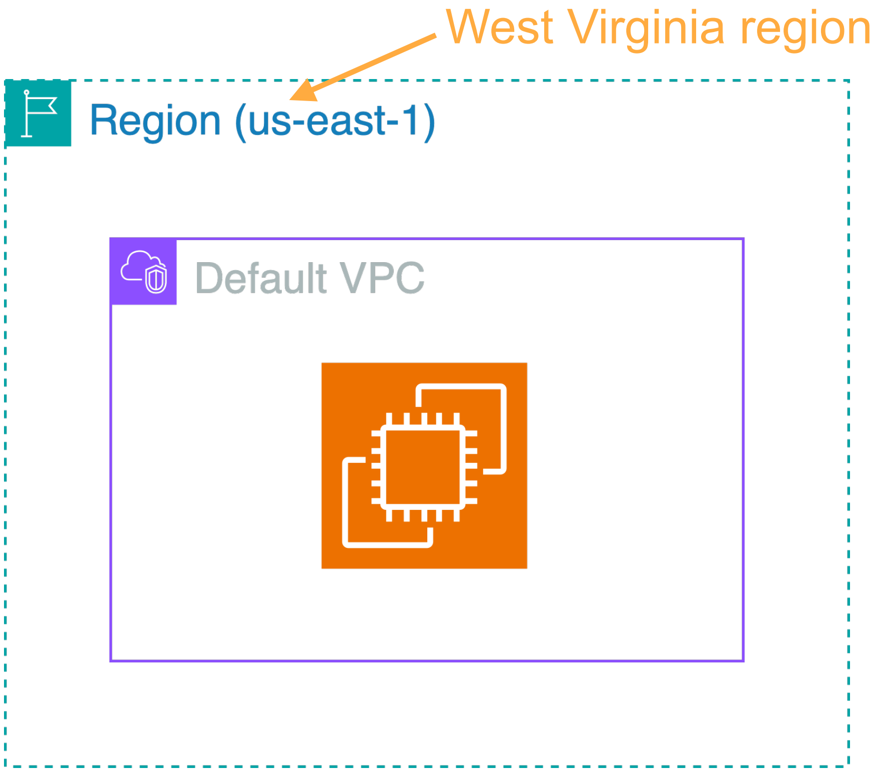
↓

You approve the execution plan

↓

Terraform applies the proposed steps

Terraform



Terraform

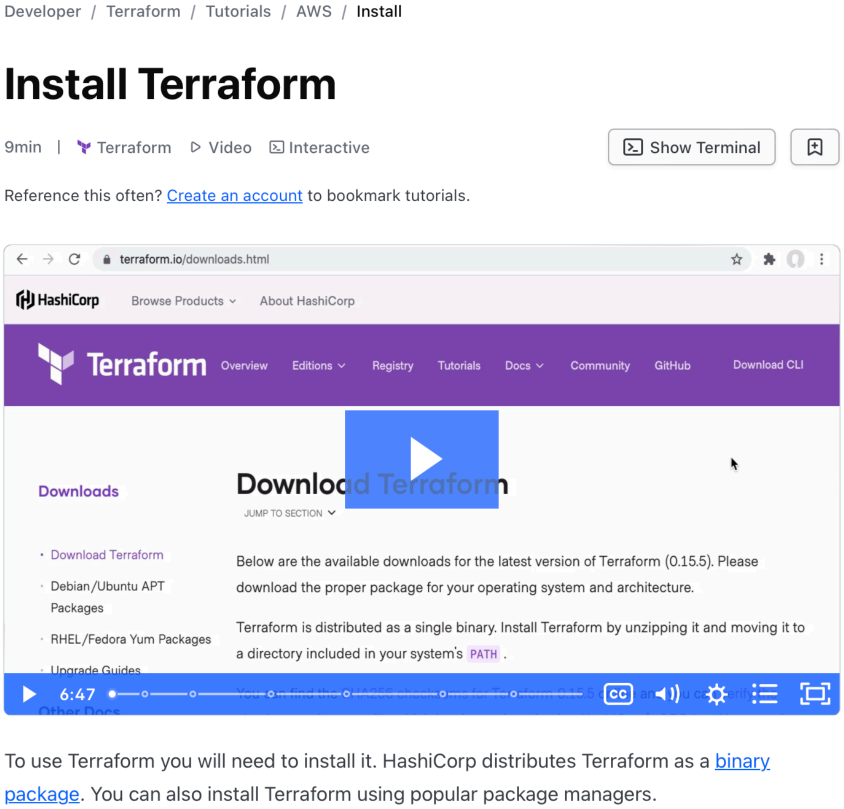
Use any IDE of your choice

↓

Install Terraform in your environment

↓

Use AWS credentials to authenticate Terraform



DeepLearning.AI

DataOps - Automation

Terraform - Defining variables and outputs

DeepLearning.AI

DataOps - Automation

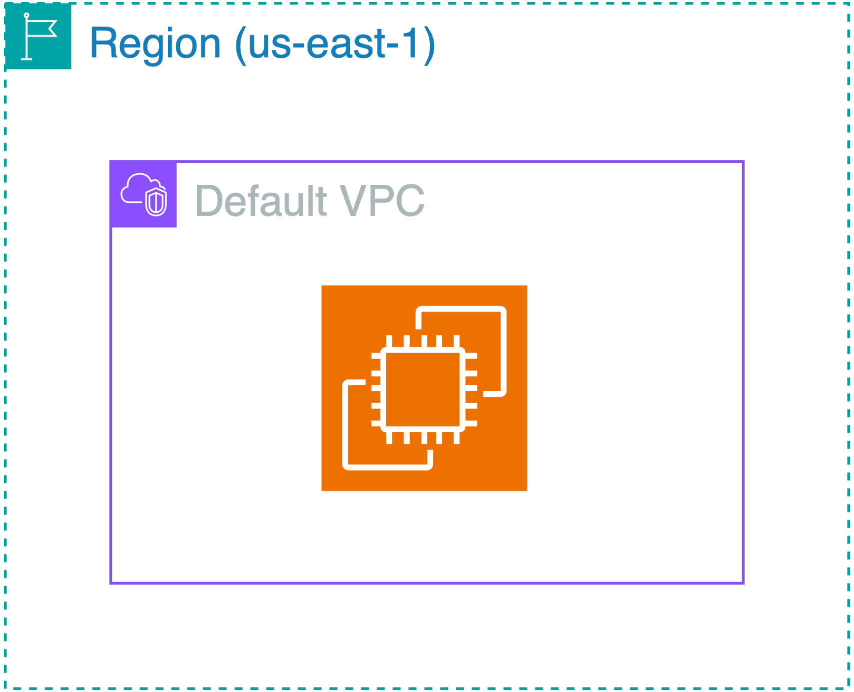
Terraform - Defining data sources and modules

Data Sources

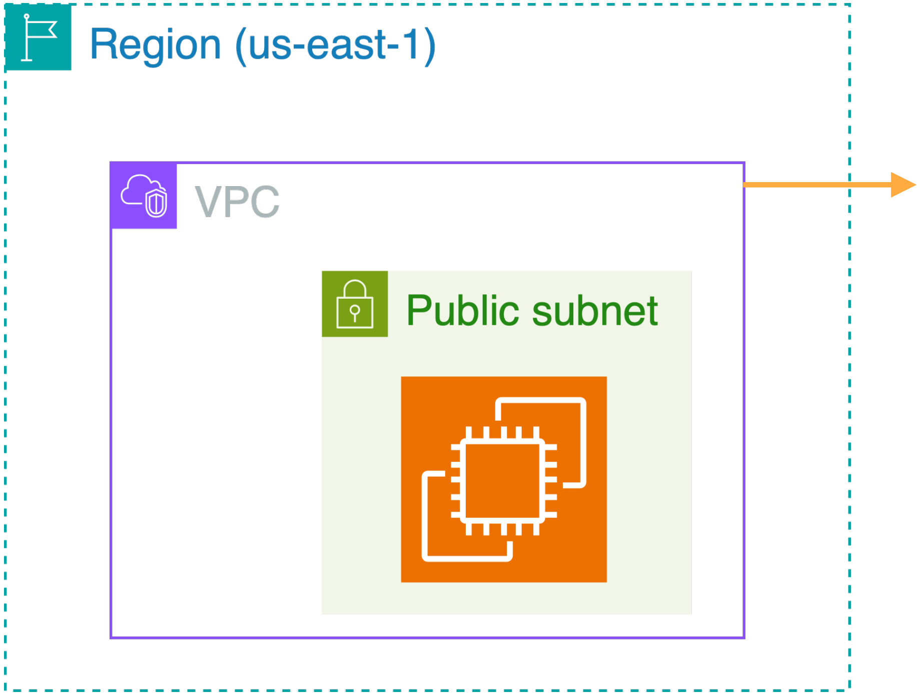
Data Sources

Data blocks to reference resources created outside Terraform or in another Terraform workspace.

Data Sources



Data Sources



VPC you have created outside the current workspace

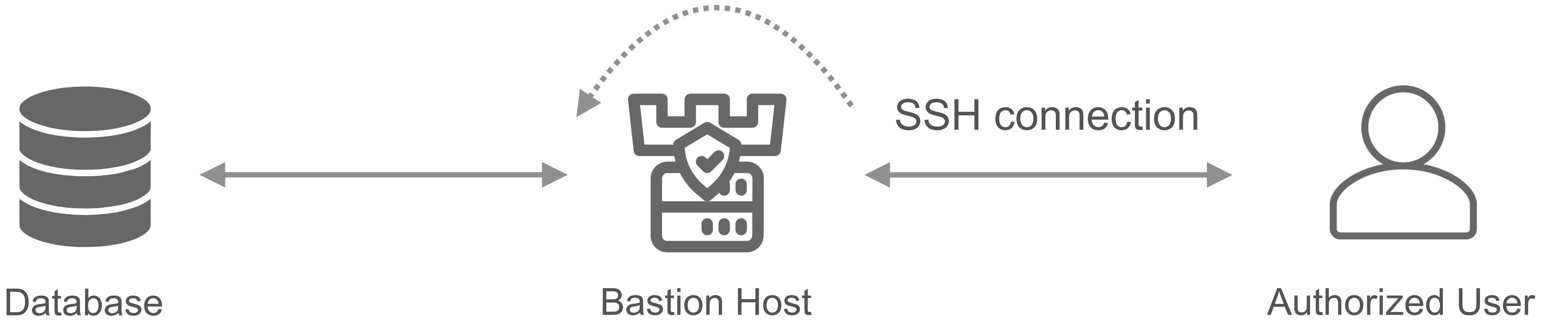
DeepLearning.AI

Lab Walkthrough

Implementing DataOps Automation with Terraform

Implementing DataOps with Terraform

- Architectural diagram of the resources
- Overview of the lab's steps

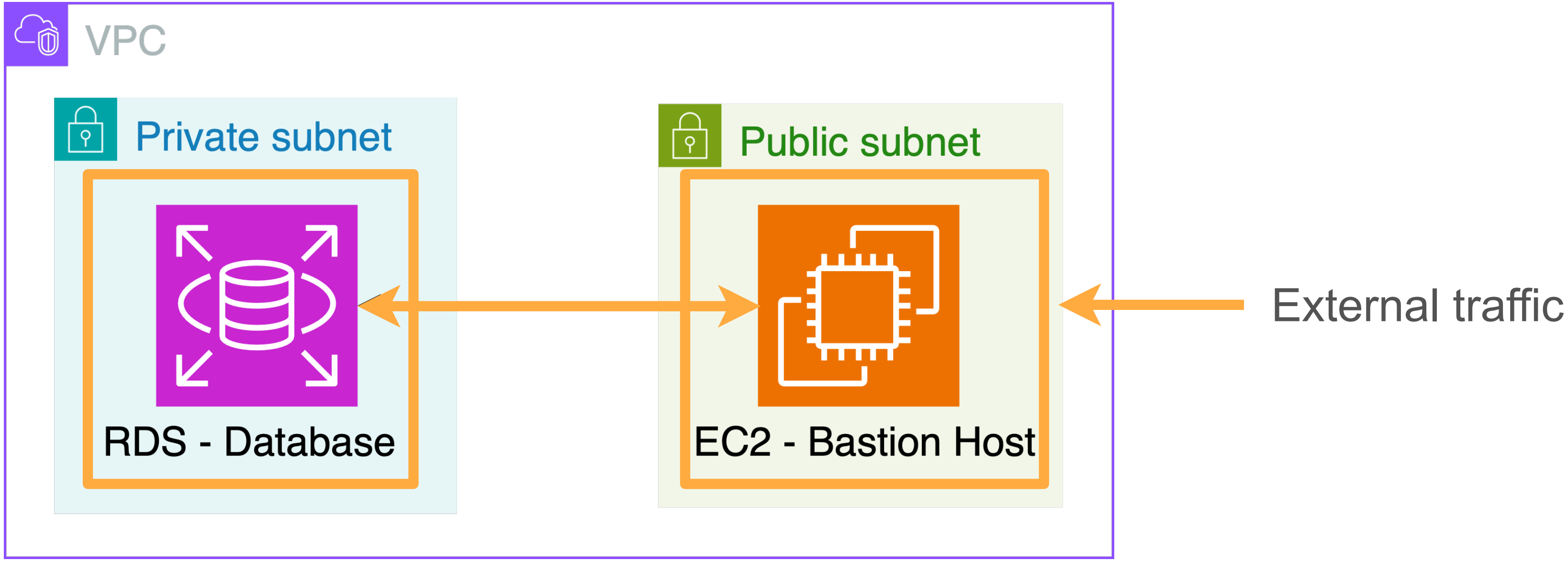


Private resource

Bridge that connects authorized users from the public internet to private resources

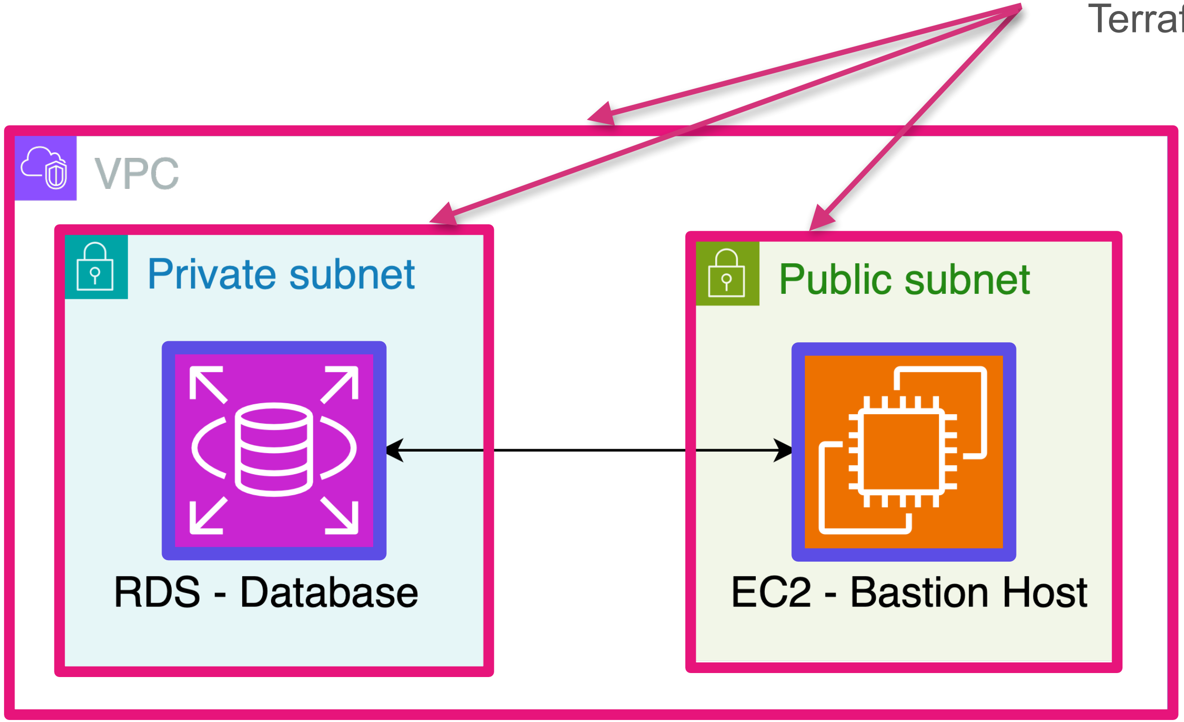
Use an SSH key to prove their identity

Architectural Diagram



Architectural Diagram

- Provided to you
- Defined as data blocks in Terraform configuration files

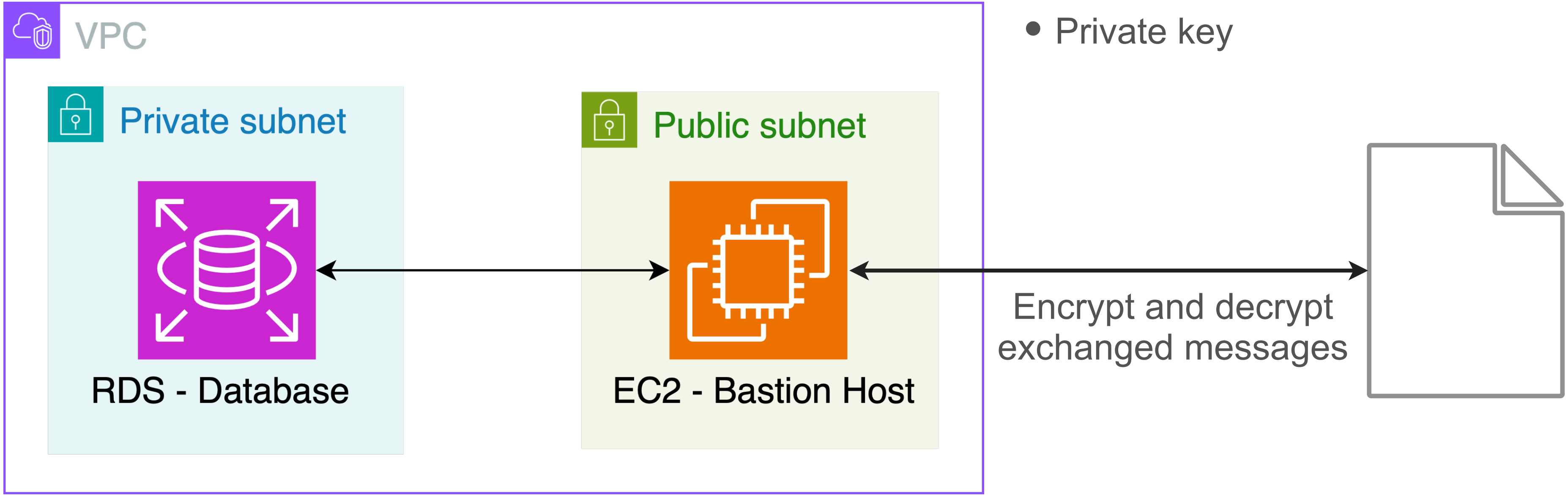


HashiCorp Terraform

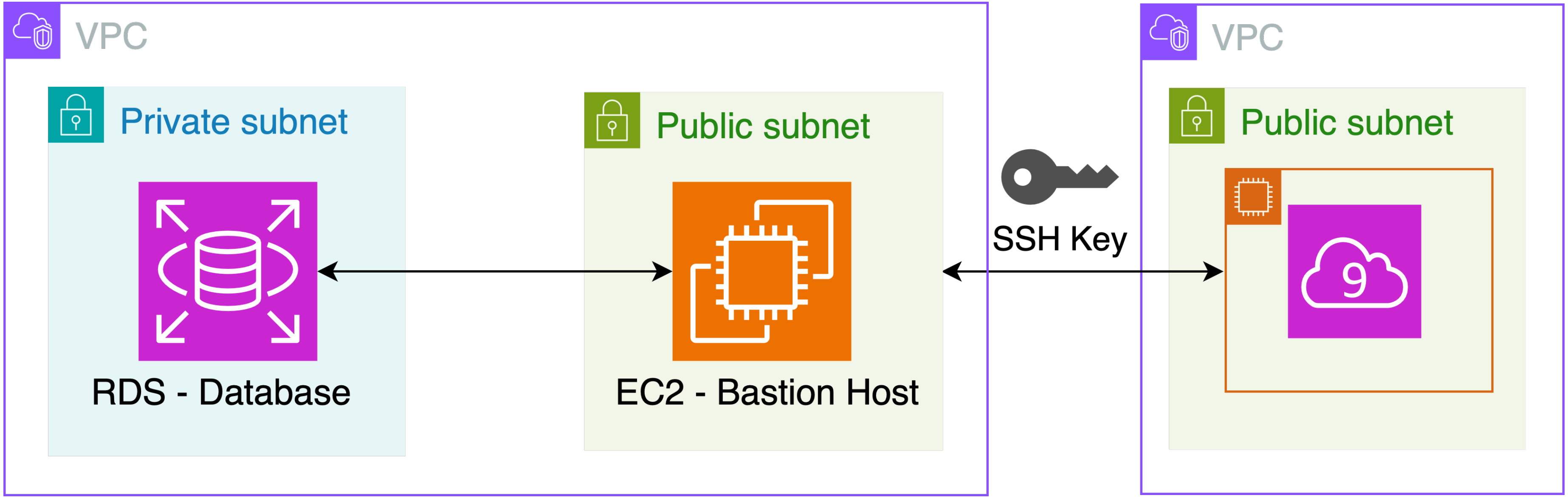
Architectural Diagram

SSH Key Pair:

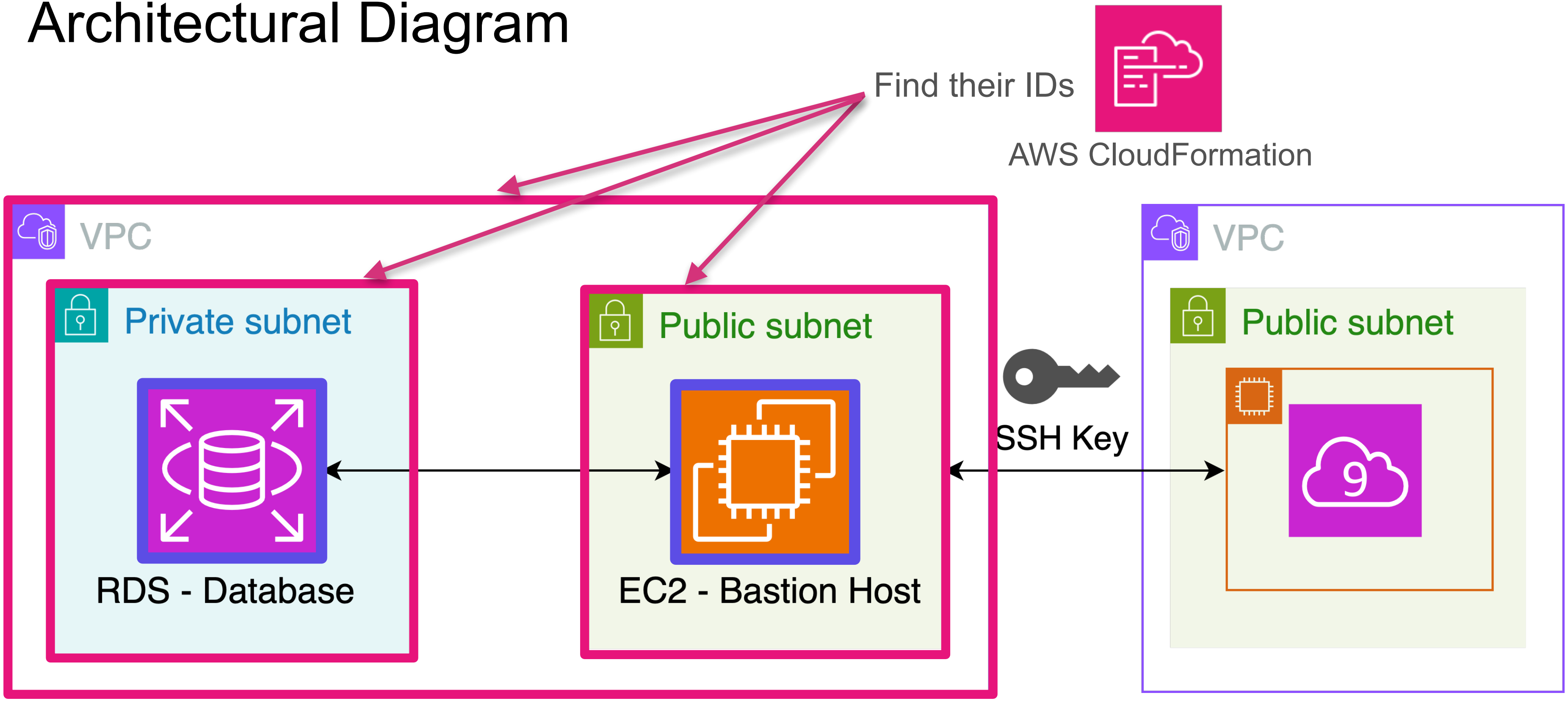
- Public key
- Private key



Architectural Diagram

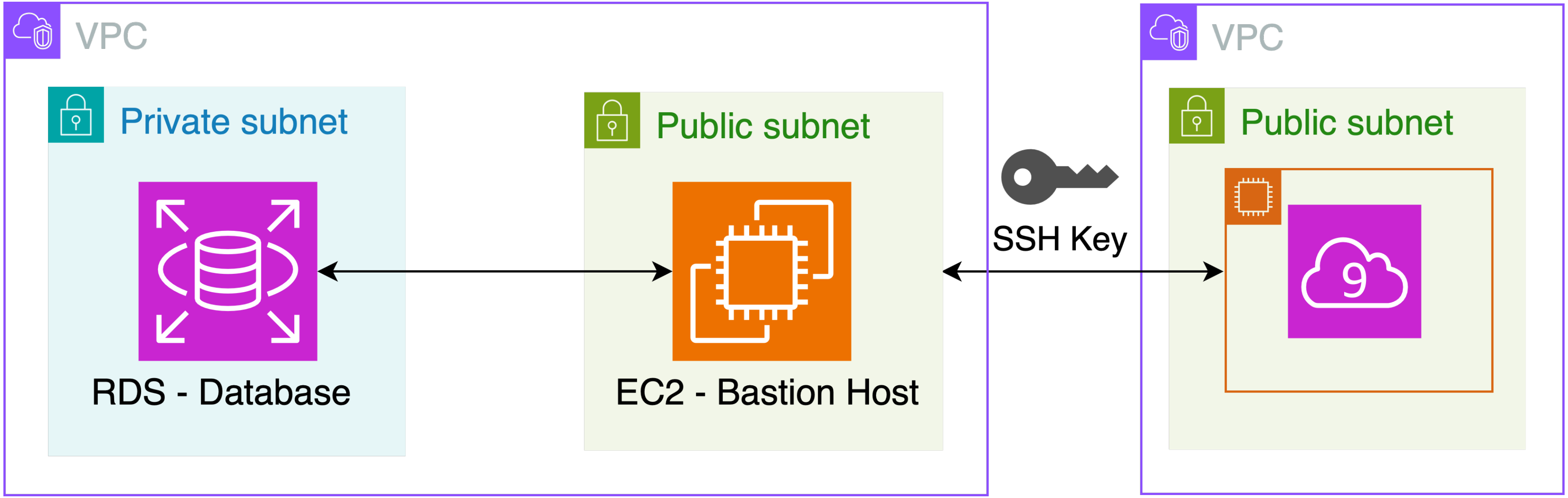


Architectural Diagram



Implementing DataOps with Terraform

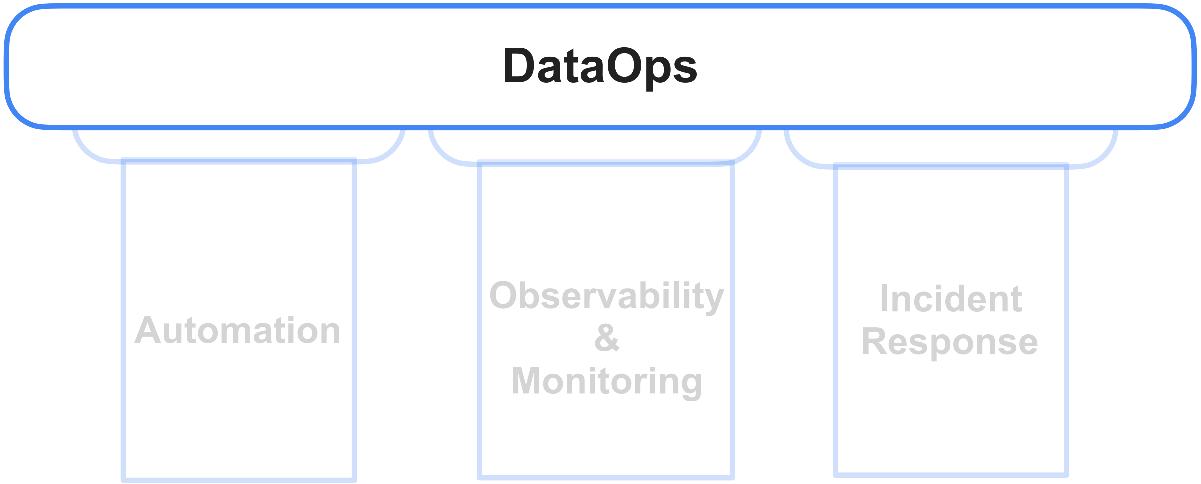
Run terraform to create your resources



DataOps - Observability & Monitoring

Data Observability

Observability and Monitoring



DevOps Observability

Advent of the Cloud

Move toward Distributed Systems

Observability tools

Gain visibility into systems' health

Metrics

CPU and RAM usage

Response time

- Quickly detect anomalies
- Identify problems
- Prevent downtime
- Ensure reliable software products

Data Observability

Monitor the health of data systems

Observability tools

Monitor the health and quality of data

Health/quality of your data

What is High Quality Data?

High Quality Data	Low Quality Data
Accurate	Inaccurate
Complete	Incomplete
Discoverable	Hard to find
Available in a timely manner	Late

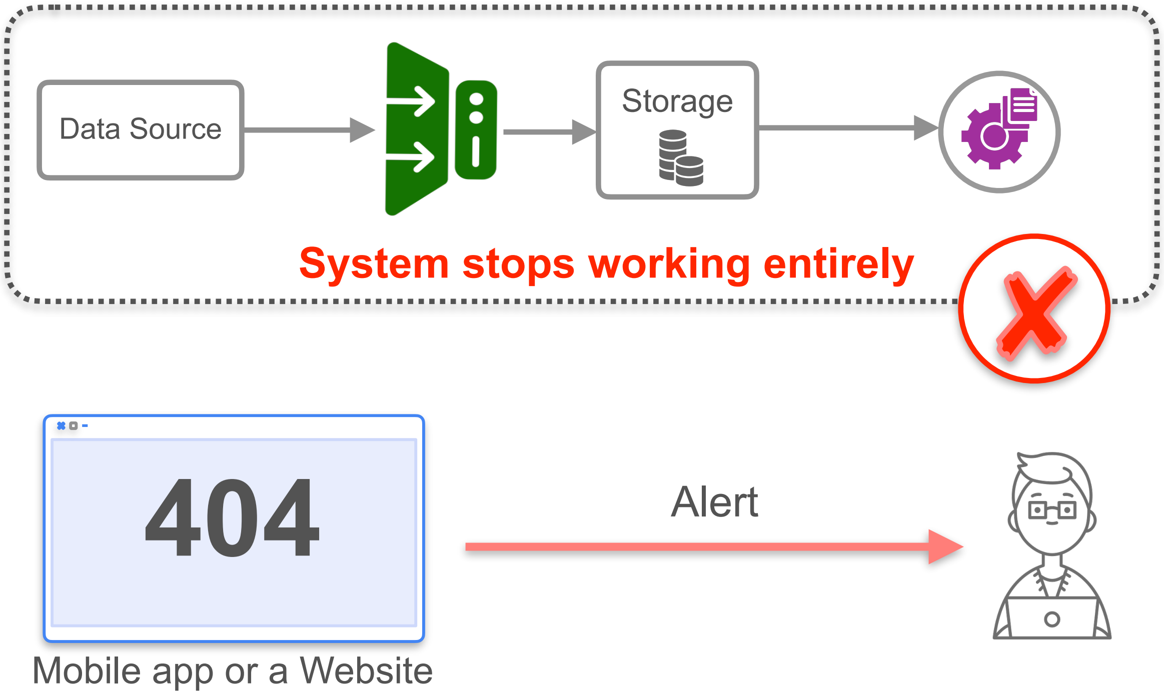
Represents exactly what stakeholders expect:

- well-defined schema
- data definitions

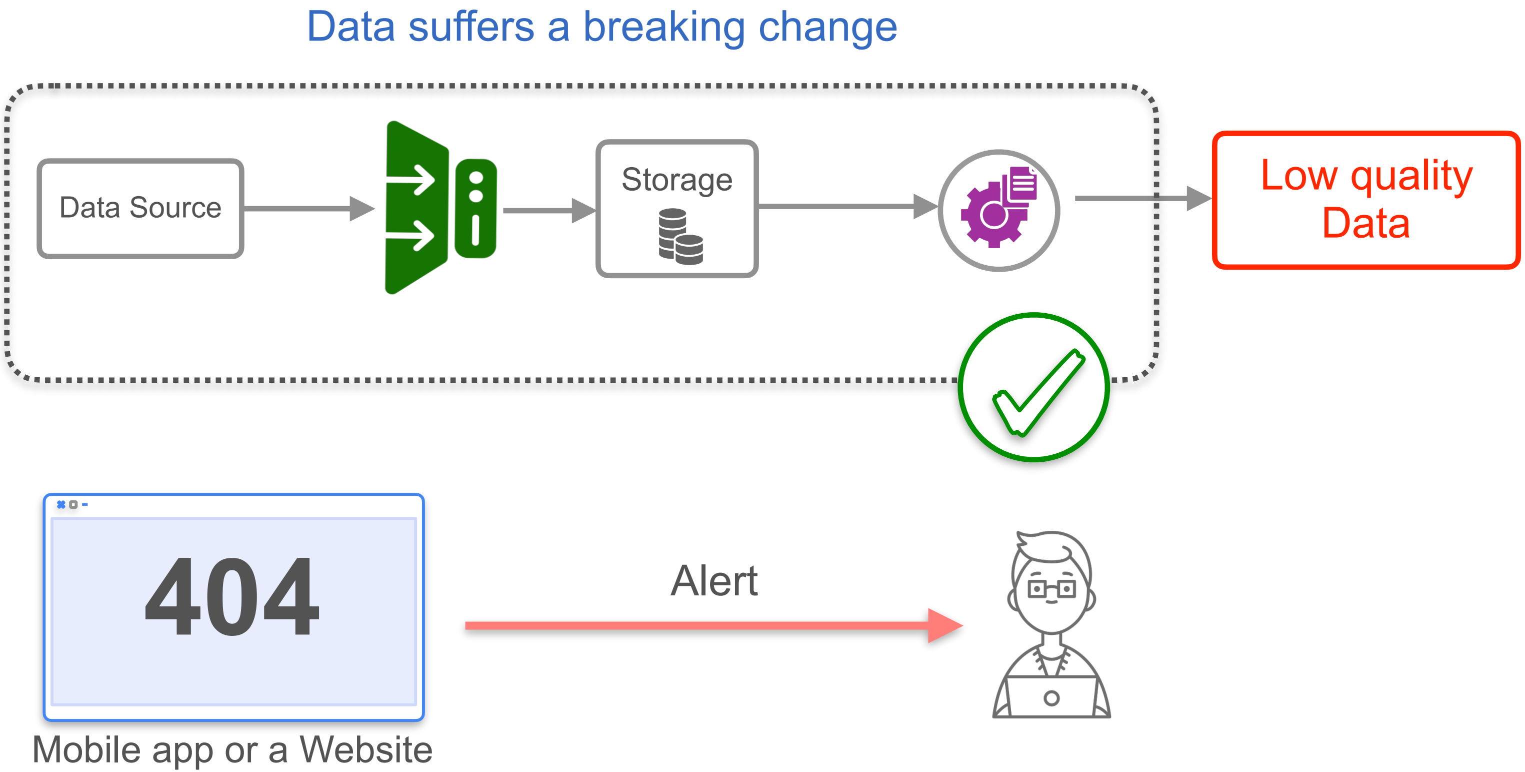
Quality of Data

Data Observability

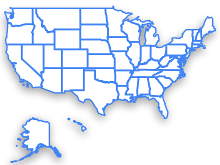
System failure is your best case scenario



Data Observability

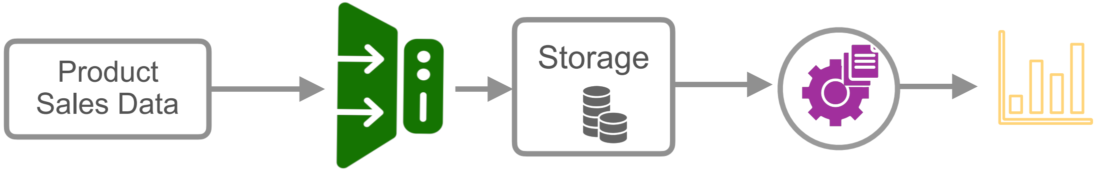


Scenario



U.S. based
company

Sell products in Europe



sale id	sale price (USD)
1	10
2	20
3	50

Total: 80

sale id	sale price (EUR)
1	9.15
2	18.31
3	45.77

Total:73.23

It looks like there was a sudden Drop in Revenue

DataOps - Observability & Monitoring

DeepLearning.AI

Barr Moses

Co-founder & CEO of Monte Carlo

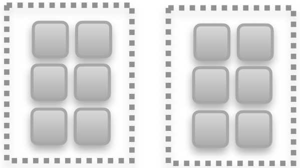
DataOps - Observability & Monitoring

DeepLearning.AI

Monitoring Data Quality

Data Quality Metrics

- **Volume**: Total number of records ingested in each batch or over some time interval



- **Null values**: Total number of null values in a table

col 1	col 2	col 3
1	10	100
2	null	null
null	50	600

- **Distribution**: Range of values in a particular column stays within some thresholds

col 1	col 2	col 3
1	10	100
2	20	300
3	50	600

- **Freshness**: The *difference* in time between *now* and the *timestamp of the most recent* record in your data

- Identify and focus on the most important metrics
- Avoid creating confusion and “alert fatigue”

Data Quality Metrics

Stakeholders

What do stakeholders care about most for this use case?

Stakeholder Requirement

Current data: no more than 24 hours old

What to monitor?

“Freshness” of the data: measure when the latest records were ingested

Data Quality Metrics

Stakeholders interested in product sales revenue

Sales

sale id	product id	purchase amount
1	2	10
2	3	20
3	4	null

Product

product id	product name	SKU
1	product 1	yk3
2	product 2	yk3
3	product 3	sj6

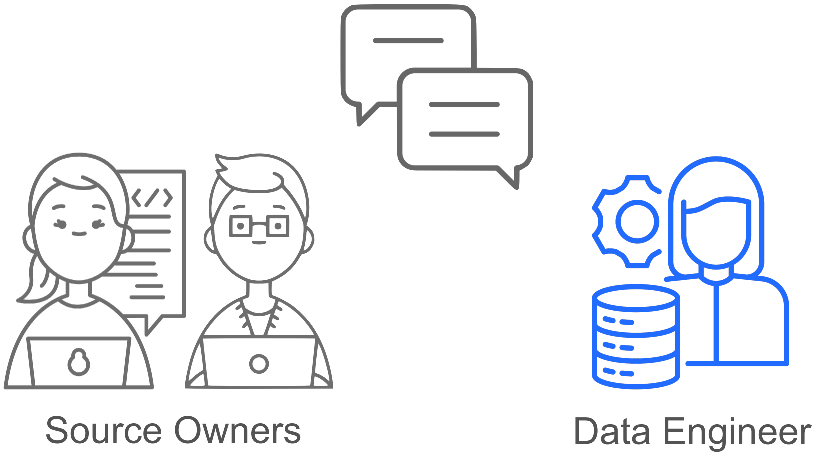
What to monitor?

- Verify that all sales records were successfully ingested
- No null values in sales record

What may be less important?

- Product SKU numbers match product descriptions
- Each customer's postal code was recorded correctly

Testing Ingested Data



- Build in checks or tests to verify that the schema and data types stay consistent.
- Identify problems as early as possible before they propagate further down your pipelines.

DeepLearning.AI

DataOps - Observability & Monitoring

Abe Gong

Co-founder & CEO of Great Expectations

DataOps - Observability & Monitoring

DeepLearning.AI

Great Expectations - Core Components

great expectations

1. Specify the data
2. Define your expectations
3. Validate your data against the expectations

Data Context → Data Sources → Expectations → Checkpoints

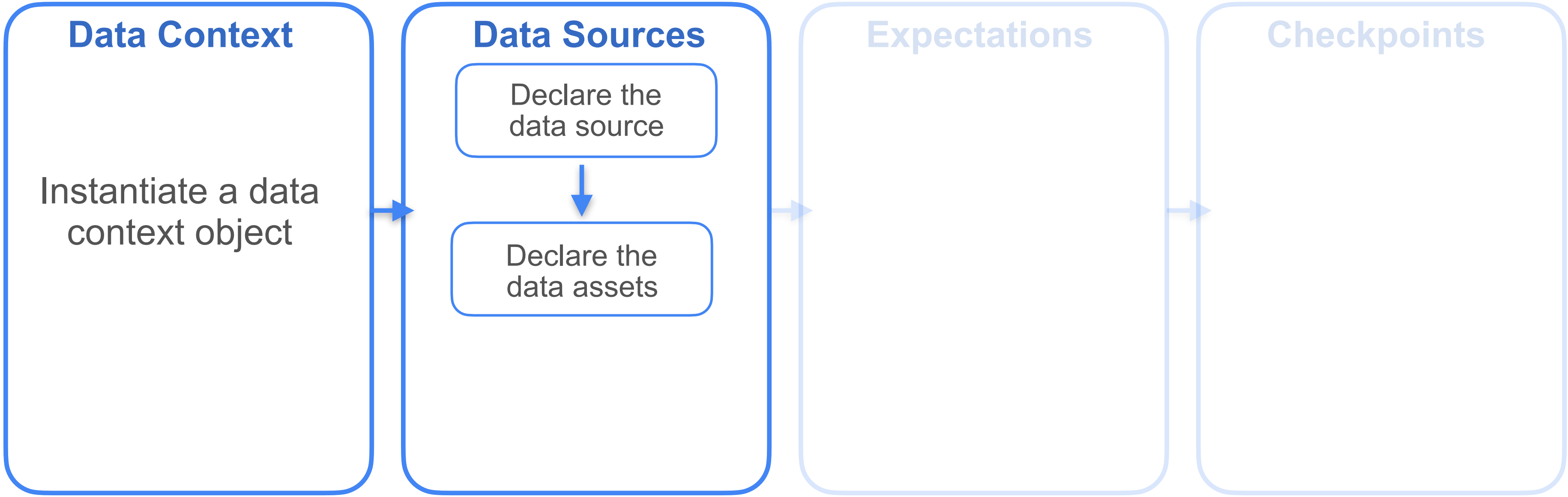
Great Expectations Workflow

- **Data Context**: entry point for the Great Expectations API
- **Great Expectations API**: classes and methods that allow you to connect data sources, create expectations and validate your data

Data Context	Data Sources	Expectations	Checkpoints
Instantiate a data context object			

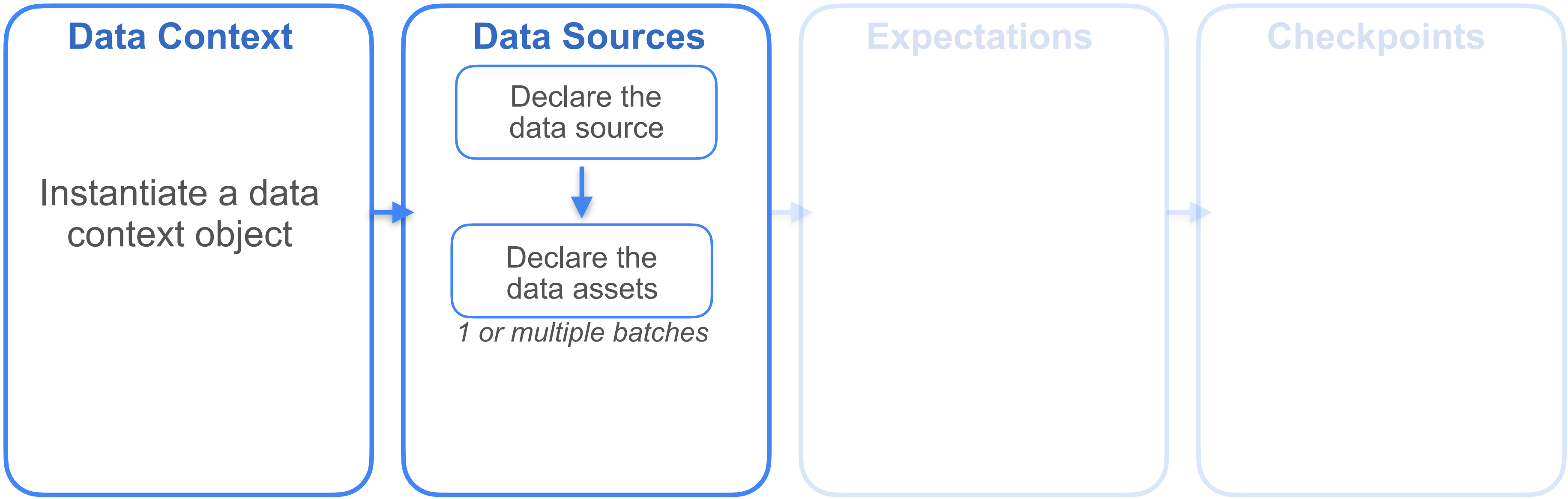
Great Expectations Workflow

- **Data source:** SQL Database, a local file system, an S3 bucket, a Pandas DataFrame
- **Data asset**: collection of records within a data source
  - Table in a SQL database
  - File in a file system
  - Join query asset
  - Collections of files matching a pattern



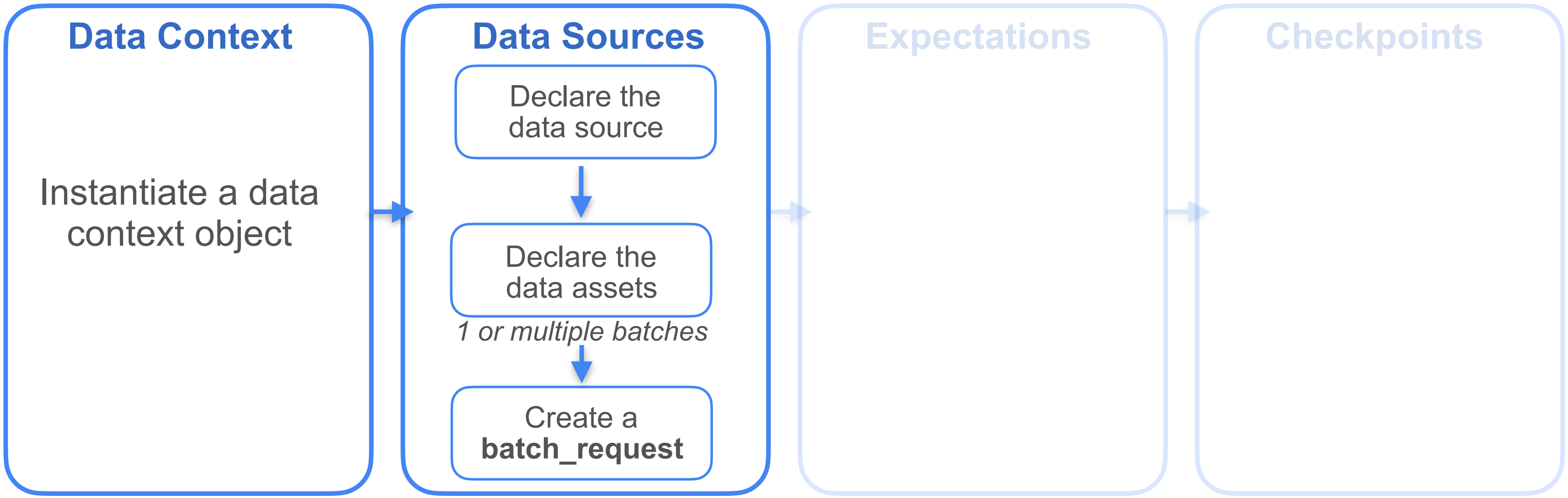
Great Expectations Workflow

- **Data source:** SQL Database, a local file system, an S3 bucket, a Pandas DataFrame
- **Data asset**: collection of records within a data source
- **Batches**: partitions from your data asset
  - partitions by date
  - partitions by column values



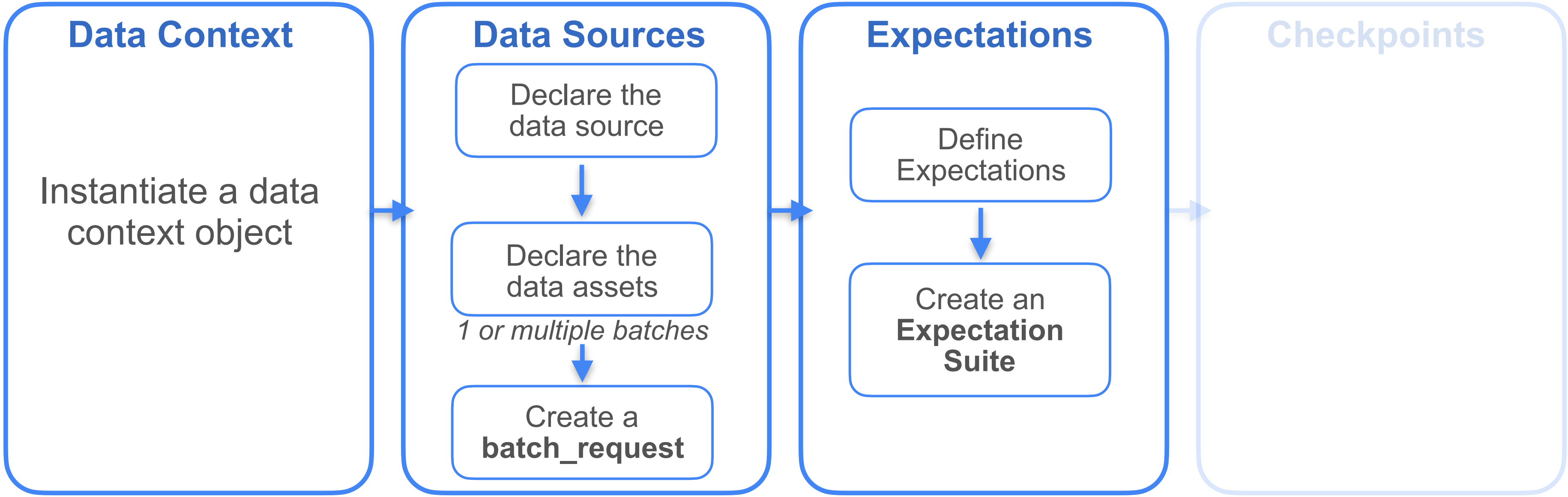
Great Expectations Workflow

- **Data source:** SQL Database, a local file system, an S3 bucket, a Pandas DataFrame
- **Data asset**: collection of records within a data source
- **Batches**: partitions from your data asset
- **Batch_request**: primary way to retrieve data from the data asset



Great Expectations Workflow

- **Expectation:** statement that you can use to verify if your data meets a certain condition
  - expect_column_min_to_be_between
  - expect_column_values_to_be_unique
  - expect_column_values_to_be_null



Great Expectations Workflow

- **Validator:** expects a batch_request and its corresponding expectation suite
  - Manual interaction
  - Or automating the validation process using a checkpoint object

Data Context

Instantiate a data context object

→

Data Sources

Declare the data source

↓

Declare the data assets

1 or multiple batches

↓

Create a **batch_request**

→

Expectations

Define Expectations

↓

Create an **expectation suite**

→

Checkpoints

- batch_request
- expectation suite

↓

Create a checkpoint

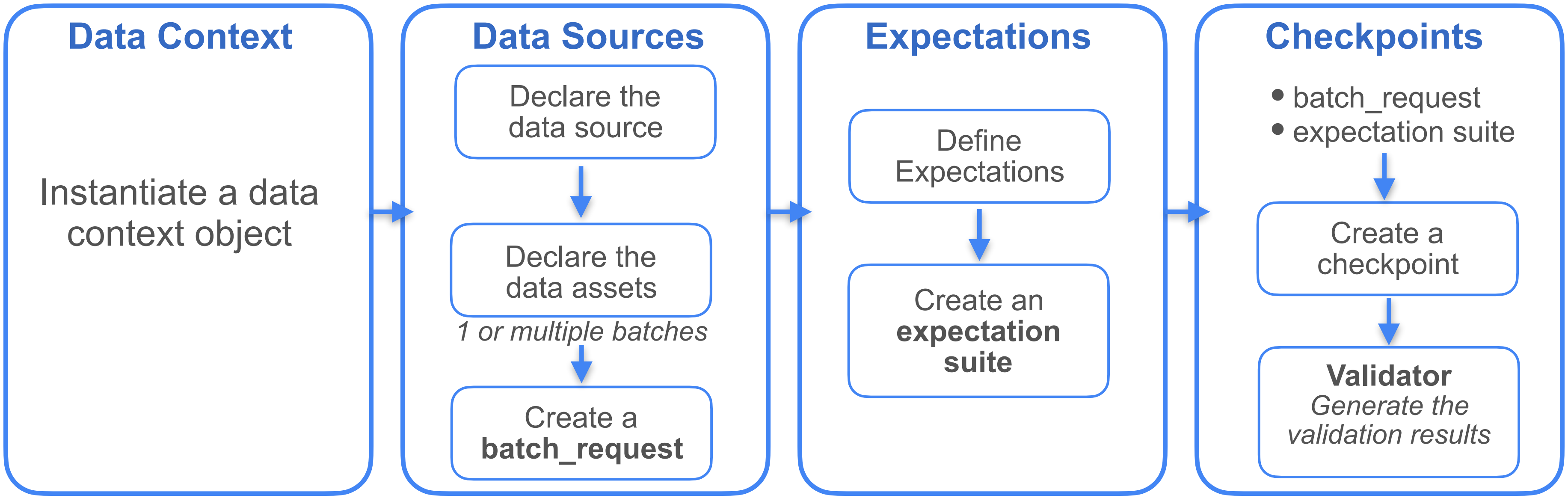
↓

Validator
Generate the validation results

Great Expectations Workflow

Metadata is saved in backend stores:

- Expectation Store
- Validation Store
- Checkpoint Store
- Data docs



DeepLearning.AI

DataOps - Observability & Monitoring

Great Expectations - Workflow Example

Workflow Example

DVD rental Database

Local postgresSQL database

payment
payment_id
customer_id
staff_id
rental_id
amount
payment_date

- payment_id contains unique IDs
- customer_id does not contain null values
- all the values in the amount column are non-negative

Data Context

Instantiate a data context object

Data Sources

Declare the data source

Declare the data assets

1 or multiple batches

Create a **batch_request**

Expectations

Define Expectations

Create an **expectation suite**

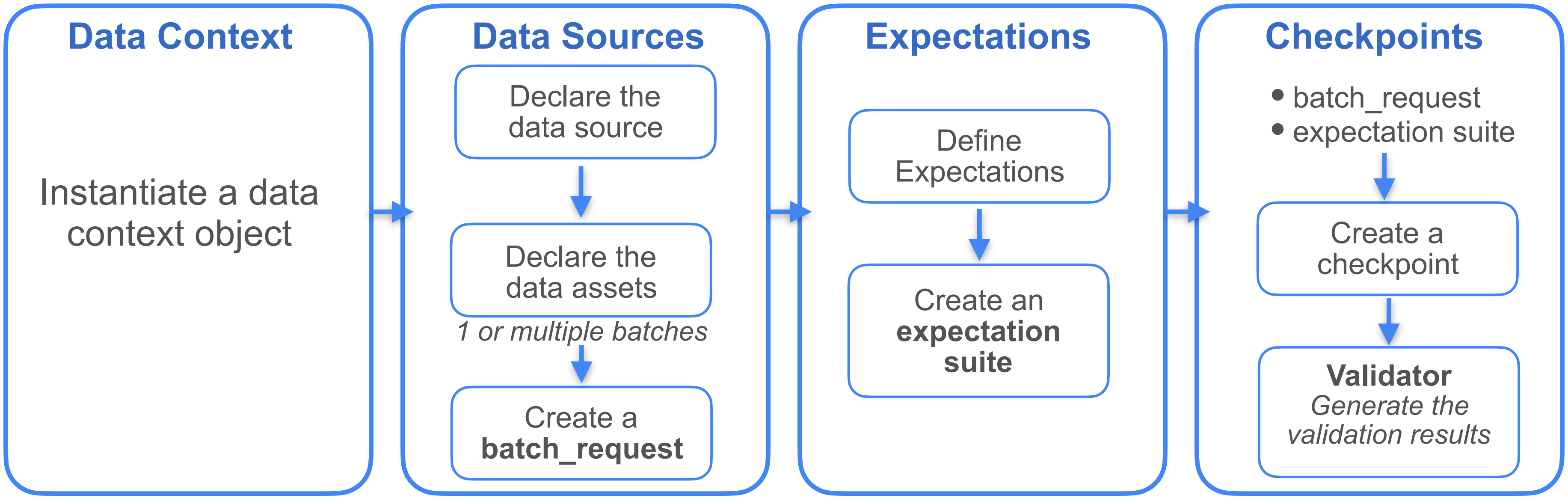
Checkpoints

- batch_request
- expectation suite

Create a checkpoint

Validator
Generate the validation results

Workflow Example



DeepLearning.AI

DataOps - Observability & Monitoring

Chad Sanderson
CEO of Gable.ai

DeepLearning.AI

DataOps - Observability & Monitoring

Amazon CloudWatch

Amazon CloudWatch

System Level Metrics:

- CPU utilization
- Disk I/O
- Network traffic
- Memory usage

Amazon CloudWatch

Made aware when metrics reflect issues

CloudWatch Alarms

System Level Metrics:

Provide a general understanding of how your resources are performing

- CPU utilization
- Disk I/O
- Network traffic
- Memory usage

Custom Metrics:

Allow you to monitor specific aspects of your application that are not covered by default system metrics

- Number of transactions processed
- Response time of an API endpoint
- Number of active users

- Define thresholds for these metrics
- Establish a baseline

  Measure performance of your system under different loads and conditions
- CloudWatch retains metrics data for up to 15 months

Common Metrics for RDS

CPU Utilization

- High value: your RDS instance might be under heavy load
- Values over 80-90%: can lead to performance bottlenecks

RAM Consumption

- High RAM consumption: can slow down performance

Disk Space

- Value consistently above 85%: you may need to delete or archive data to free up some space

Database Connections

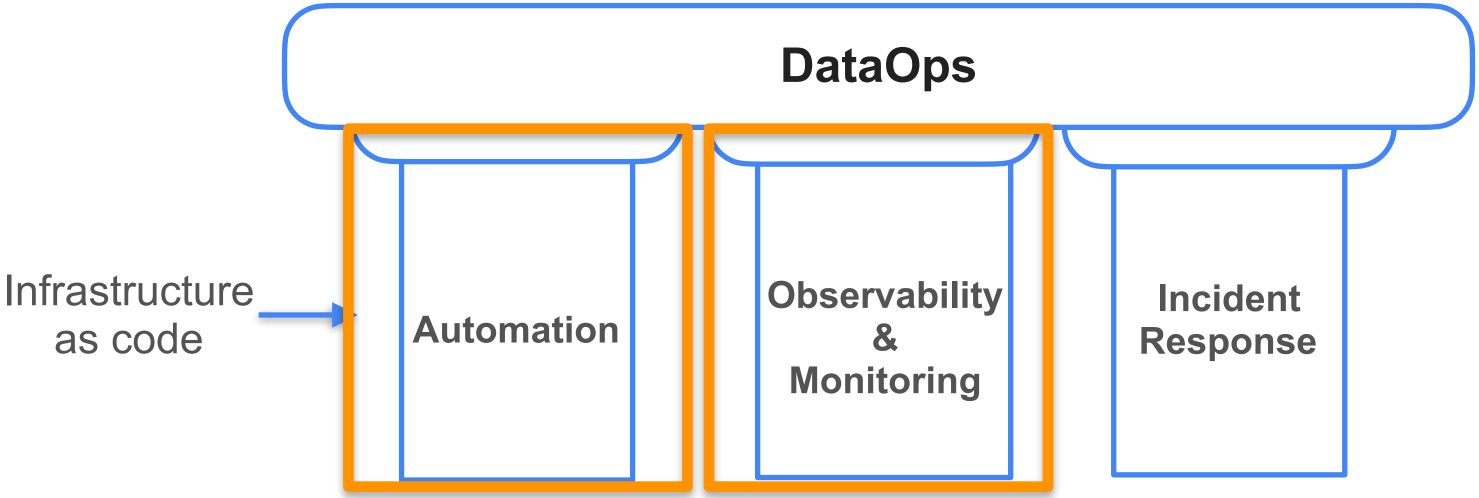
- The number of active connections to your database
- Number of connections approaching the maximum limit: can lead to connection errors and application failures

DeepLearning.AI

DataOps

Summary

DataOps Summary



Ensure that you're delivering value for stakeholders.

DataOps Summary

Chris Bergh

Barr Moses

Abe Gong

Chad Sanderson

In This Week's Labs

HashiCorp Terraform

- A cloud agnostic tool for developing and maintaining your data infrastructure as a codebase
- Infrastructure as Code tools help make deployments consistent and repeatable

Amazon CloudWatch

great expectations